Department of Computer Science

Course Name & No.: Cloud Computing 1501438

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Ch 4- Cloud Computing Model

Assignment 4

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Task 1: NIST Cloud Definition
Summarize the NIST definition of cloud computing in your own words (2–3 sentences). Include the three main components of the model.

Task 2: Essential Characteristics
List the five essential characteristics of cloud computing according to NIST, and give a brief example of each.

4. Characteristic:
Example:

2. Characteristic:
Example:

3. Characteristic:
Example:

4. Characteristic:
Example:

5. Characteristic:
Example:

Task 3: Cloud Actors and Roles
Match each cloud actor with its primary role in the NIST model.

Answer	Actor	Role
	1. Cloud Consumer	A. Provides connectivity and transport for cloud services.
	2. Cloud Provider	B. Uses cloud services and may be billed for usage.
	3. Cloud Broker	C. Evaluates cloud services for security, privacy, and performance.
	4. Cloud Auditor	D. Aggregates, customizes, or manages cloud services.
	5. Cloud Carrier	E. Owns and manages cloud infrastructure and offers services.

Task 4: Deployment Models
Identify the correct deployment model for each scenario below:

- Public Cloud
- Private Cloud
- Community Cloud
- Hybrid Cloud

Scenarios:

4. A university sets up a cloud only for its students and faculty.
   Model:

2. A hospital shares cloud resources with other hospitals in the same city for medical data collaboration.
   Model:

3. A startup uses AWS for its website and keeps sensitive customer data in its own data center.
   Model:

4. Netflix streams movies globally using shared infrastructure.
   Model:

Task 5: Why Use a Cloud Broker?
List three reasons why an organization might use a cloud broker, as explained in the chapter.

Task 1: NIST Cloud Definition

Cloud computing is a model that allows users to access computing resources like servers, storage, and applications over the internet on demand, without needing to manage physical infrastructure. It is designed to be scalable, flexible, and pay-as-you-go. The model includes three main components: **essential characteristics, service models, and deployment models**.

Task 2: Essential Characteristics

1. **Characteristic:** On-Demand Self-Service
   Example: A user can create a virtual machine instantly without contacting the provider.
2. **Characteristic:** Broad Network Access
   Example: Accessing cloud applications like Google Drive from a phone or laptop anywhere.
3. **Characteristic:** Resource Pooling
   Example: Multiple users share the same physical servers in a data center.
4. **Characteristic:** Rapid Elasticity
   Example: A website automatically scales up during high traffic and scales down afterward.
5. **Characteristic:** Measured Service
   Example: A company is billed only for the storage or bandwidth it actually uses.

Task 3: Cloud Actors and Roles

Answer Actor Role

Answer	Actor	Role
B	1. Cloud Consumer	Uses cloud services and may be billed for usage.
E	2. Cloud Provider	Owns and manages cloud infrastructure and offers services.
D	3. Cloud Broker	Aggregates, customizes, or manages cloud services.
C	4. Cloud Auditor	Evaluates cloud services for security, privacy, and performance.

Task 4: Deployment Models

1. A university sets up a cloud only for its students and faculty.
   Model: Private Cloud
2. A hospital shares cloud resources with other hospitals in the same city.
   Model: Community Cloud
3. A startup uses AWS but keeps sensitive data locally.
   Model: Hybrid Cloud
4. Netflix streams movies globally using shared infrastructure.
   Model: Public Cloud

Task 5: Why Use a Cloud Broker?

1. To **combine multiple cloud services** from different providers into one solution.
2. To **optimize cost and performance** by selecting the best services.
3. To **simplify management** of cloud services through a single point of control.